

PAPER_667: UQFF Black Hole Stability Mathematical Proofs

Subtitle: Four-proof mathematical chain demonstrating UQFF extends black hole Hawking evaporation timescales by $\sim 30\times$. **Module:** UQFFBlackHoleStabilityProofs

Session: Session 172

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Abstract

We provide four sequential mathematical proofs demonstrating that UQFF vacuum field interactions extend black hole evaporation timescales by a combined factor of ~ 30 . Each proof addresses a distinct UQFF mechanism.

Proof 1 — Negentropic Confinement

$$\tau' = \frac{\tau_{Hawking}}{1 - f_{TRZ}} \approx 1.11 \tau_{Hawking}$$

$f_{TRZ} = 0.1$ suppresses the rate of pair annihilation at the horizon.

Proof 2 — Aether/SCm Gradient Barrier

Energy barrier: $E_{barrier} = k_B T_H \cdot (\rho_{SCm}/\rho_{UA})$

$$\tau'' = \tau' \cdot \frac{\rho_{UA}}{\rho_{SCm}} \approx 10 \tau'$$

Proof 3 — U_m String Anchoring

$$\tau_{UQFF} = \tau'' \cdot \exp\left(\frac{U_m}{k_B T_H}\right) \approx 2.718 \tau''$$

Proof 4 — Combined

$$\tau_{UQFF} = \tau_{Hawking} \cdot \frac{1}{1 - f_{TRZ}} \cdot \frac{\rho_{UA}}{\rho_{SCm}} \cdot \exp\left(\frac{U_m}{k_B T_H}\right)$$

$$\text{Factor} = 1.11 \times 10 \times 2.718 \approx \mathbf{30}$$

Numerical Verification

Mass	τ_{Hawking}	τ_{UQFF}	Factor
1 M $\square$	2.1×10^{74} yr	$\sim 6 \times 10^{75}$ yr	30×
Sgr A*	$\square$ t _H	$\square\square$ t _H	30×

C++ Module

UQFFBlackHoleStabilityProofs.h / .cpp — Session 172 CP4 #251 — UQFFBlackHoleStabilityProofsCalculator

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